

JOBSHEET TJKT

Mail Server

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TJKT XI-3

Mata Pelajaran : Administrasi Sistem Jaringan

Kompetensi : Mengkonfigurasi Mail Server

Unjuk Kerja/Produk : Instalasi dan Konfigurasi Mail Server Linux Debian

TUJUAN

Siswa dapat melakukan instalasi dan konfigurasi Mail Server pada Linux Debian Jessie

KESELAMATAN KERJA

1. Siswa wajib menggunakan wearpack saat praktikum
2. Menggunakan alat dan bahan sesuai fungsinya
3. Bacalah petunjuk penggunaan alat dan bahan dengan baik dan benar
4. Bila mendapatkan masalah atau kesulitan tanyakan pada guru
5. Tidak bercanda atau makan dan minum pada saat praktikum
6. Rapihan dan kembalikan alat dan bahan sesuai tempatnya

DASAR TEORI

Webmail (Web-based Mail) adalah aplikasi email client yang berbasis web. Jika sebelumnya kamu pernah melihat atau menggunakan layanan email seperti Yahoo dan Gmail. Mereka menggunakan webmail sebagai email client untuk setiap penggunanya, misalnya Gmail yang bisa diakses di alamat www.gmail.com.

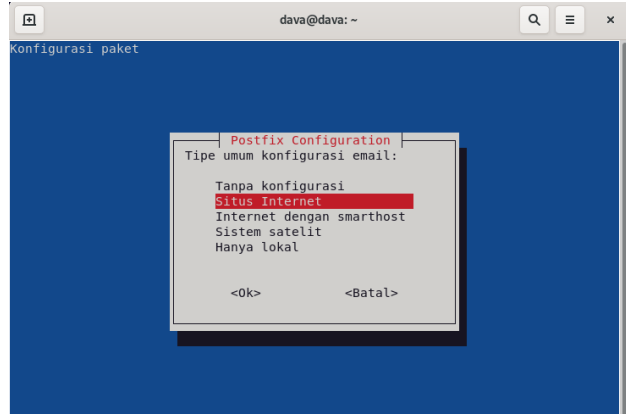
ALAT DAN BAHAN

1. PC/Laptop
2. CD Debian 8.2 DVD 1 dan CD Debian 8.2 DVD 2
3. Software VirtualBox ver 5.2
4. Internet
5. SSH
6. Software PUTTY

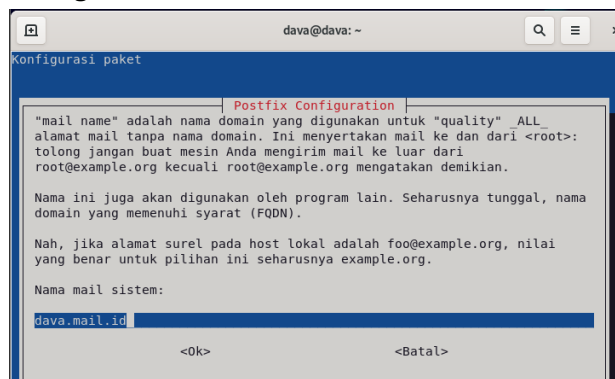
1. Install package

```
root@dava:~# apt install postfix dovecot-imapd dovecot-pop3d
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  dovecot-core
Suggested packages:
  dovecot-gssapi dovecot-ldap dovecot-lmtpd dovecot-lucene
  dovecot-managesieved dovecot-mysql dovecot-pgsql dovecot-sieve dovecot-solr
  dovecot-sqlite dovecot-submissiond ntp procmail postfix-mysql postfix-pgsql
  postfix-ldap postfix-pcre postfix-lmdb postfix-sqlite resolvconf postfix-cdb
  postfix-doc
The following NEW packages will be installed:
  dovecot-core dovecot-imapd dovecot-pop3d postfix
9 upgraded, 4 newly installed, 0 to remove and 336 not upgraded.
Need to get 8.791 kB of archives.
After this operation, 19,3 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://kartolo.sby.datautama.net.id/debian bullseye/main i386 dovecot-cor
i386 1:2.3.13+dfsg1-2+deb11u1 [4.505 kB]
7% [1 dovecot-core 738 kB/4.505 kB 16%]
```

2. Pilih package



3. Konfigurasi domain



4. Edit **/etc/postfix/main.cf** dan tambahkan **home_mailbox = Maildir/** pada baris paling bawah.

```
inet_interfaces = all
inet_protocols = all
```

```
home_mailbox = Maildir/
```

5. Buat directory baru

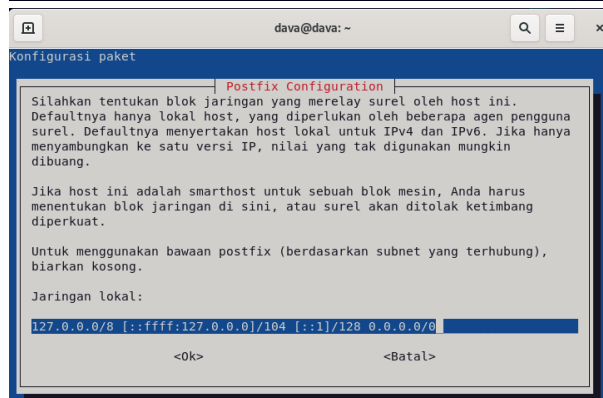
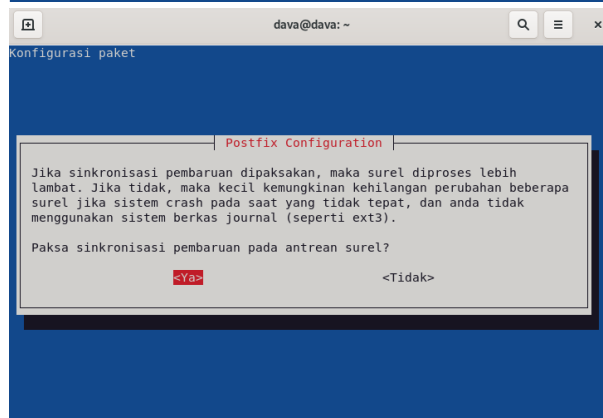
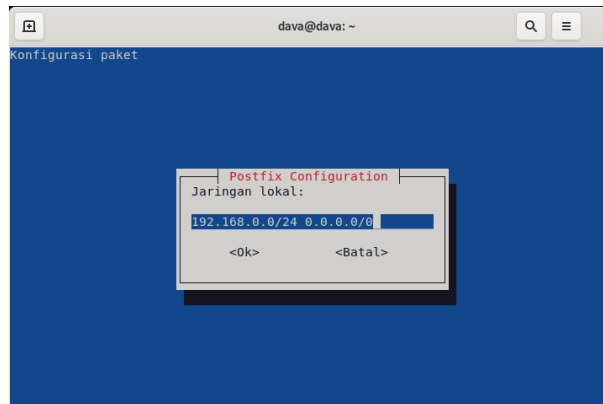
```
root@dava:~# maildirmake.dovecot /etc/skel/Maildir
root@dava:~#
```

6. Setelah itu masukan perintah

dpkg-reconfigure postfix, jika tidak bisa tambahkan 'sudo'

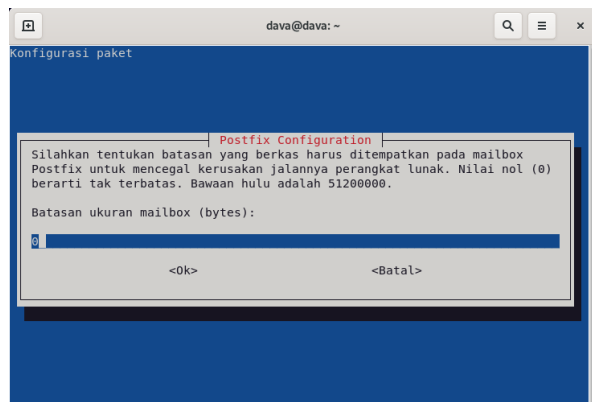
Lalu ikuti dan sesuaikan saja beberapa konfigurasi dibawah ini

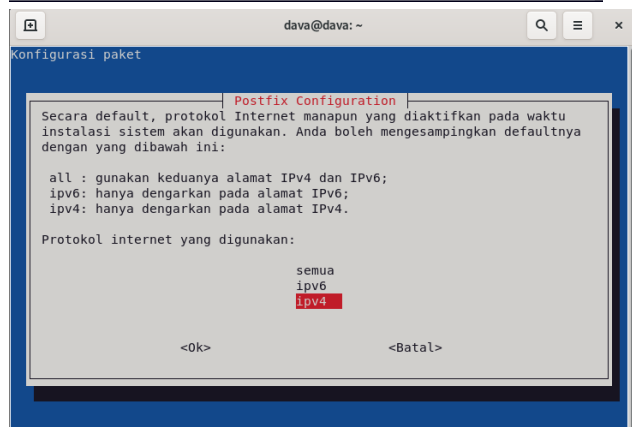
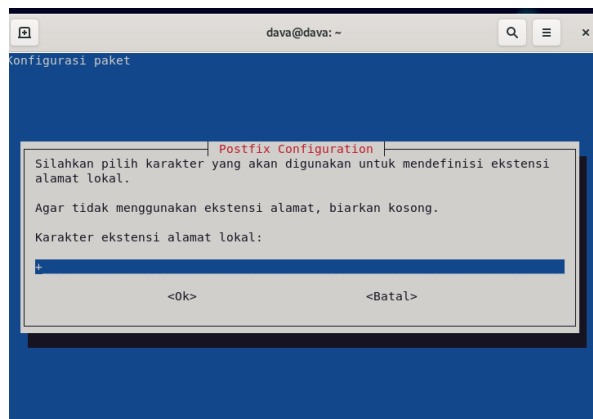




tambahkan network ip server dan

0.0.0.0/0 agar postfix dapat menerima request dari semua traffic





7. Setelah selesai reconfigure, restart servicenya

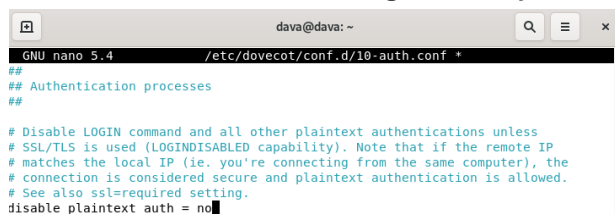
```
root@dava:~# systemctl restart postfix
root@dava:~#
```

8. Edit file konfigurasi /etc/dovecot/dovecot.conf uncomment dan edit pada line ke 30 'Listen = *'



9. Edit file konfigurasi /etc/dovecot/conf.d/10-auth.conf

Uncomment line ke 10 dan ganti dari yes ke no



10. Edit file konfigurasi /etc/dovecot/conf.d/10-mail.conf

```
# mail_location = maildir:~/Maildir
```

```
#mail_location = mbox:~/mail:INBOX=/var/mail/%u
```

```
root@dava:~# systemctl restart dovecot
root@dava:~#
```

```
root@dava:~# sudo adduser satu
Adding user `satu' ...
Adding new group `satu' (1002) ...
Adding new user `satu' (1002) with group `satu' ...
Creating home directory `/home/satu' ...
Copying files from `/etc/skel' ...
```

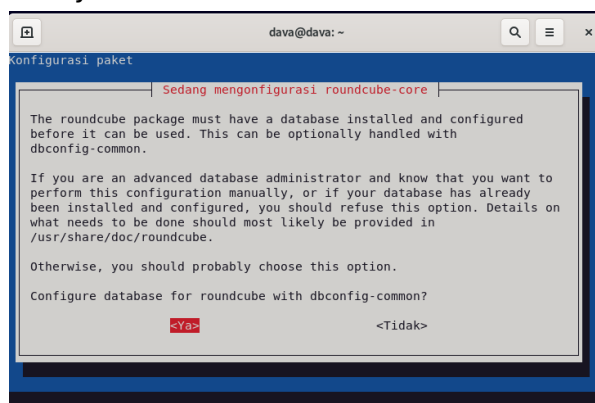
```
root@dava:~# systemctl restart postfix dovecot
root@dava:~#
```

```
dava@dava:~$
dbconfig-mysql | dbconfig-pgsql | dbconfig-sqlite3 | dbconfig-no-thanks
gawk-doc libimltdm-perl libnet-daemon-perl libssl-statement-perl
libipc-sharedcache-perl libjs-requires libjs-jquery-ui-docs mailx
mariadb-test netcat-openbsd roundcube-plugins php-net-ldap3 php-crypt-gpg
php-mkopinsky-zxcvbn-php

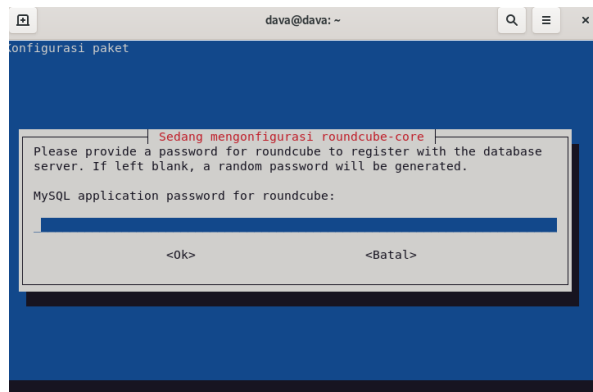
The following NEW packages will be installed:
dbconfig-common fonts-glyphicons-halflings galera-4 gawk libaio1
libapache2-mod-php libapache2-mod-php7.4 libcgi-fast-perl libcgi-pm-perl
libcginif-infiles-perl libdbd-mariadb-perl libdbi-perl libfcgi-bin
libfcgi-perl libfcgi0ldbl libhtml-template-perl libjs-bootstrap
libjs-bootstrap4 libjs-codemirror libjs-jquery libjs-jquery-minicolors
libjs-jquery-ui libjs-jstestrunner libjs-popper.js libjs-sizzle
libmariadb3 libonig5 libjsiseqv2 libterm-readkey-perl mariadb-client-10.5
mariadb-client-core-10.5 mariadb-common mariadb-server mariadb-server-10.5
mariadb-server-core-10.5 mysql-common node-jquery php-auth-sasl php-common
php-gd-cli php-gd-im php-gd-libpng php-gd-mt php-gd-mt2 php-gd-mt3 php-mysql
php-net-sieve php-net-smtp php-net-socket php-pear php-pspell php-xml
php7.4-cgi php7.4-common php7.4-gd php7.4-intl php7.4-json php7.4-mbstring
php7.4-mysql php7.4-opcache php7.4-pspell php7.4-readline php7.4-xml
roundcube roundcube-core roundcube-mysql rsync socat

0 upgraded, 67 newly installed, 0 to remove and 333 not upgraded.
Need to get 32.3 MB of archives.
After this operation, 210 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Pilih yes untuk membuat database secara otomatis



Masukan password database roundcube



Edit file /etc/roundcube/config.inc.php

*Isikan default host dengan nama domain mail server**

*Ganti smtp server dengan nama domain mail server**

*Ganti smtp port dari 587 ke 25**

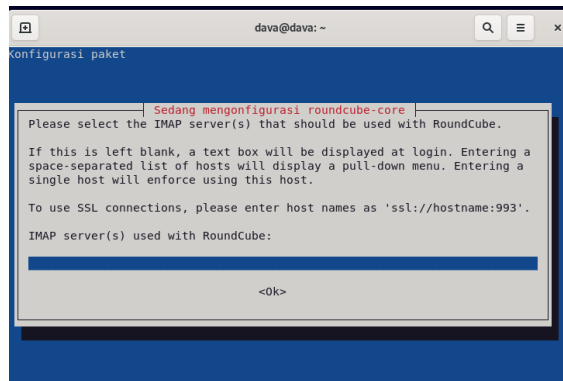
*Kosongkan value dari smtp user**

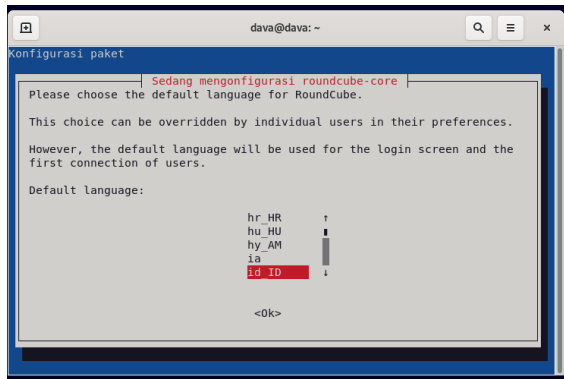
*Kosongkan value dari smtp password**

```
// %s - domain name after the '@' from e-mail address provided at login screen
// For example %n = mail.domain.tld, %t = domain.tld
$config['default_host'] = 'dava.mail.com';
```

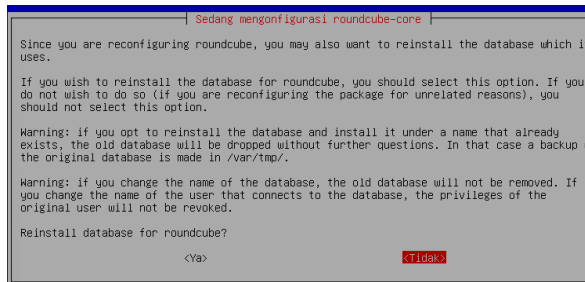
16. Reconfigure roundcube

Kosongkan karena tidak menggunakan tls

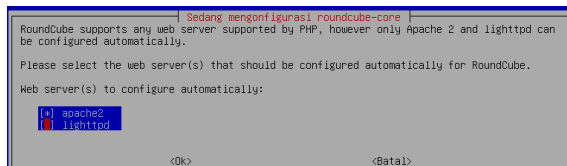




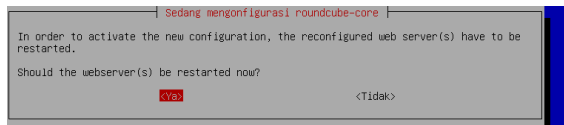
Pilih no karena database telah dibuat



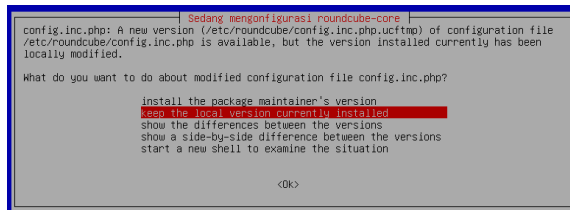
Check pada apache2 dan uncheck pada lighttpd



Pilih yes untuk merestart web server



Keep local version jika tidak ingin merubah versi roundcube ke yang lebih baru



Edit apache config untuk memasukan konfigurasi tambahan dari roundcube ke
apache config “nano /etc/apache2/apache2.conf

Tambahkan pada baris paling bawah ‘Include /etc/roundcube/apache.conf


```

GNU nano 5.4 /etc/apache2/apache2.conf *
/FilesMatch>

# The following directives define some format nicknames for use with
# a CustomLog directive.

# These deviate from the Common Log Format definitions in that they use %D
# (the actual bytes sent including headers) instead of %b (the size of the
# requested file), because the latter makes it impossible to detect partial
# requests.

# Note that the use of %I (Forwarded-For) instead of %h is not recommended.
# Use mod_remoteip instead.

logFormat "%v:%p %h %l %u %t \"%r\" %s %D \"%{Referer}i\" \"%{User-Agent}i\"" vhost_combined
logFormat "%h %l %u %t \"%r\" %s %D \"%{Referer}i\" \"%{User-Agent}i\"" combined
logFormat "%h %l %u %t \"%r\" %s %D" common
logFormat "%{Referer}i -> %D" referer
logFormat "%{User-Agent}i" agent

# Include of directories ignores editors' and dpkg's backup files,
# see README.Debian for details.

# Include generic snippets of statements
includeOptional conf-enabled/*.conf

# Include the virtual host configurations:
includeOptional sites-enabled/*.conf

# vim: syntax=apache ts=4 sw=4 sts=4 sr noet

include /etc/roundcube/apache.conf_

```

17. Pindah ke direktori /etc/apache2/sites-available

```
root@dava:/etc/apache2/sites-available# cd /etc/apache2/sites-available
```

18. Buat file mail.conf

```
root@dava:/etc/apache2/sites-available# touch mail.conf
```

19. Edit mail.conf dan tambahkan lalu sesuaikan konfigurasi berikut

```

GNU nano 5.4 mail.conf *
<VirtualHost *:80>
    ServerName tkjt.dava.local
    DocumentRoot /usr/share/roundcube
</VirtualHost>

```

20. Install bind9

```

root@dava:~# systemctl restart dovecot postfix
root@dava:~# apt install bind9
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bind9-dnswriter bind9-libs bind9-utils dns-root-data python3-ply
Suggested packages:
  bind-doc resolvconf python-ply-doc
The following NEW packages will be installed:
  bind9 bind9-utils dns-root-data python3-ply
The following packages will be upgraded:
  bind9-dnswriter bind9-libs
2 upgraded, 4 newly installed, 0 to remove and 114 not upgraded.
Need to get 2.986 kB of archives.
After this operation, 2.451 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bullseye/main 1:9.16.50-1-deb11u2 [415 kB]
Get:2 http://deb.debian.org/debian bullseye/main 1:9.16.50-1-deb11u2 [1.513 kB]
Get:3 http://deb.debian.org/debian bullseye/main 1:9.11-4 [65,5 kB]
Get:4 http://deb.debian.org/debian bullseye/main 1:9.16.50-1-deb11u2 [343 kB]
Get:5 http://deb.debian.org/debian bullseye/main 1:9.16.50-1-deb11u2 [5.864 B]
Get:6 http://deb.debian.org/debian bullseye/main 1:9.16.50-1-deb11u2 [514 kB]
debconf: unable to initialize frontend: Dialog
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: falling back to frontend: Teletype
Unpacking bind9 (1:9.16.50-1-deb11u2) ...
Unpacking bind9-libs (1:9.16.50-1-deb11u2) ...
Unpacking bind9-utils (1:9.11-4) ...
Unpacking dns-root-data (1:2024041801-deb11u1) ...
Setting up bind9 (1:9.16.50-1-deb11u2) ...
Setting up bind9-libs (1:9.16.50-1-deb11u2) ...
Setting up bind9-utils (1:9.11-4) ...
Setting up dns-root-data (1:2024041801-deb11u1) ...

```

21. Edit dan sesuaikan pada file '/etc/bind/named.conf.local'

Ganti db.127 dengan db.192

```

dava@server0504: ~
GNU nano 5.4 /etc/bind/named.conf.local
//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "mail.tkjt0504.net" {
    type master;
    file "/etc/bind/db.mail";
};

zone "27.3.10.in-addr.arpa" {
    type master;
    file "/etc/bind/db.192";
};

```

22. Cp db.local menjadi db.mail lalu konfigurasi dan sesuaikan isi filenya

Editnya di **db.mail**

```
dava@server0504: ~  
GNU nano 5.4 /etc/bind/db.mail  
; BIND data file for local loopback interface  
;  
$TTL 604800  
@ IN SOA mail.tkj0504.net. root.mail.tkj0504.net. (  
    2      ; Serial  
    604800 ; Refresh  
    86400  ; Retry  
    2419200 ; Expire  
    604800 ) ; Negative Cache TTL  
;  
@ IN NS mail.tkj0504.net.  
@ IN A 10.3.27.227  
ns1 IN A 10.3.27.227
```

23. Cp db.127 menjadi db.192

Editnya di **db.192**

```
dava@server0504: ~  
GNU nano 5.4 /etc/bind/db.192  
; BIND reverse data file for local loopback interface  
;  
$TTL 604800  
@ IN SOA mail.tkj0504.net. root.mail.tkj0504.net. (  
    1      ; Serial  
    604800 ; Refresh  
    86400  ; Retry  
    2419200 ; Expire  
    604800 ) ; Negative Cache TTL  
;  
@ IN NS mail.tkj0504.net.  
A IN PTR 10.3.27.227  
3 IN PTR ns1
```

24. Edit resolv.conf

```
dava@dava: ~  
GNU nano 5.4  
nameserver 192.168.27.1  
nameserver 8.8.8.8  
nameserver 8.8.4.4
```

25. Edit /etc/hosts, tambahkan ip address dan domain pada baris paling bawah

```
dava@dava: ~  
GNU nano 5.4 /etc/hosts *  
127.0.0.1 localhost  
127.0.1.1 dava  
  
# The following lines are desirable for IPv6 capable hosts  
::1 localhost ip6-localhost ip6-loopback  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters  
  
192.168.27.1 tjkt.dava.local
```

26. Disable apache default dan enable email config

```
root@dava:~# a2dssite 000-default.conf  
-bash: a2dssite: perintah tidak ditemukan  
root@dava:~# a2dissite 000-default.conf  
Site 000-default disabled.  
To activate the new configuration, you need to run:  
systemctl reload apache2  
root@dava:~# a2ensite mail.conf  
Enabling site mail.  
To activate the new configuration, you need to run:  
systemctl reload apache2  
root@dava:~# systemctl reload apache2  
root@dava:~# a2ensite mail.conf  
Site mail already enabled  
root@dava:~# _
```

27. Tes menggunakan telnet

```

root@dava:~# telnet tjkt.dava.local 25
Trying 192.168.27.1...
Connected to tjkt.dava.local.
Escape character is '^]'.
220 dava.localdomain ESMTP Postfix (Debian/GNU)
mail from:satu@tjkt.dava.local
250 2.1.0 Ok
rcpt to:dava.alkhoeri@gmail.com
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Subject:woi
hello world
.
250 2.0.0 Ok: queued as 8DA915FFC3
quit
221 2.0.0 Bye
Connection closed by foreign host.
root@dava:~# |

```

28. Tes di web

